

Sara Zhiliu He . Jarvis Consulting

I earned a Master's degree in Systems Design Engineering (Artificial Intelligence and Machine Learning Specialization) from the University of Waterloo, where I built great foundation for software engineering and AI. Eager to expand my skill set, I completed a software engineering diploma program at BrainStation, gaining hands-on experience in full-stack web development. I am an energetic software developer with professional experience as a Software Engineer at Jarvis and volunteer experience as a full-stack developer at Skill4Skill, driven by a passion for learning and applying new technologies to solve real-world problems and build innovative solutions.

Skills

Proficient: Java, Spring Boot, Linux/Bash, RDBMS/SQL, JavaScript/TypeScript, React, Python, Agile/Scrum, Git

Competent: Angular, Node.js/Express, MongoDB, CI/CD, Docker, Pandas/NumPy

Familiar: AI/Machine Learning, TensorFlow, PyTorch, Scikit-Learn, Figma

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_SarazhiliuHe

Linux Cluster Monitoring Agent [GitHub]: Built the Jarvis Linux Cluster Monitoring Agent to assist the Linux Cluster Administration (LCA) team in managing a multi-node Rocky Linux environment. Implemented a Bash Script to collect static system information, such as CPU architecture, memory, and disk specs, executed once on each node. Automated the collection of real-time resource usage data, such as CPU, memory, and disk usage, using a Bash script triggered by a Cron job every minute. Deployed a PostgreSQL database in a Docker container to store the collected data in a portable environment. Used Git and GitHub for source code management and version control.

Java Grep App [GitHub]: Developed Grep App, a Java application that mimics the Linux grep command to search for lines matching a regular expression across a directory of text files. Leveraged Java 8 Lambda and Stream APIs for efficient and readable data processing. Utilized Maven for project build and dependency management. Dockerized the application for seamless distribution and deployment.

Python Data Analytics [GitHub]: Extracted and loaded raw retail data into a PostgreSQL Data Warehouse (OLAP) provisioned locally with Docker. Conducted in-depth sales and customer behavior analysis in Jupyter Notebook using Python, Pandas, and NumPy. Delivered insights and visualizations via GitHub to support targeted marketing strategy.

Highlighted Projects

MasterMind [GitHub]: Developed and maintained a web application for educational content creators using Next.js and TypeScript, and created reusable UI components to enhance consistency and maintainability. Designed and optimized PostgreSQL databases using Prisma ORM for seamless and efficient data interactions. Implemented security best practices, including role-based access control, and integrated secure authentication with Clerk. Deployed the application on Vercel, ensuring fast load times and global availability.

InStock: Built a responsive warehouse inventory management website using HTML, CSS, React, and JavaScript, adhering to functional and visual design requirements. Developed and managed APIs with Node.js, Express.js, and Knex.js, optimizing database interactions with MySQL. Handled version control and team collaboration using Git and GitHub, utilized Jira for project management, and participated in code reviews to ensure high-quality development and on-time delivery.

Professional Experiences

Software Developer, Jarvis (APR 2025-present): Collaborated within an Agile team to deliver diverse software development projects across multiple sprints. Gained hands-on experience in technologies such as Linux/Bash scripting, SQL, Java, and Python, adapting quickly to new technical challenges. Worked with Google Cloud Platform (GCP) to provision virtual machines and access company infrastructure via remote desktop. Used Git for version control and Docker for containerization to build, test, and maintain scalable applications.

Full Stack Software Developer, Skill4Skill (MAR 2025-present): Collaborated with cross-functional teams to develop Skill4Skill, a web platform where individuals can exchange expertise. Built a responsive and accessible frontend in Next.js, ensuring alignment with design specifications and style guides, and worked closely with designers to refine UI/UX.

Designed MongoDB schemas and developed RESTful APIs with Express.js, and integrated backend APIs with frontend components for seamless feature delivery. Maintained documentation for codebases, workflows, and API specifications. Participated in Agile development using Jira for sprint planning, user story tracking, and backlog management.

Research Assistant, University of Waterloo (MAY 2024-DEC 2024): Researched and developed supply chain optimization models with Python, Pandas, Numpy and Gurobi. Documented findings in analytical reports, communicated issues and solutions effectively with the supervising professor, and refined the paper based on feedback.

Education

University of Waterloo (2023-2024), Master of Engineering, Systems Design Engineering - GPA: 85/100

South China University of Technology (2019-2023), Bachelor of Engineering, Logistics Engineering - GPA: 92/100
- First-class Academic Scholarship - Student Science and Technology Innovation Competition Scholarship - PG Logistics Scholarship

Miscellaneous

- BrainStation Software Engineering Diploma (2024)
- Balsillie School of International Affairs Technology Governance Summer School (2024)
- First prize in the National Contest in Big Data Analysis and Mining
- Third Prize in the China Undergraduate Mathematical Contest in Modeling
- Chinese Classical Dance (Captain of Dance Team at South China University of Technology)
- Member of Campus Volunteer Service Team